
Cerebro

White Paper v2.2 - Executive Summary

From Architecture Thesis to Evidence-Bounded Capability

Cerebro Cyber Solutions

Version v2.2 executive summary

Date 14 August 2026

Evidence current as of 14 August 2026.

Decision-ready update with source-built, internally proven, operating and externally validated state kept separate.

Document control

Field	Value
Document	Cerebro White Paper v2.2 - Executive Summary
Version	2.2
Evidence freeze	Merged main as of 14 August 2026; exact commit pin recorded in the repository review notes.
Continuation chain	v1.3.3-public + Supplement 1 -> v2.0 -> v2.1 -> this v2.2 release
Purpose	Provide a concise, claim-aligned summary of the Version 2.2 white paper for technology, research and partner readers.
Public-claim rule	Repository evidence is dated. Live runtime claims require a fresh owning probe.

STATUS VOCABULARY / Built means merged code or a recorded artifact. Internally proven means a bounded test or dated receipt passed. Operating internally requires an observed owning runtime path. Externally validated requires independent customer or market evidence. Operator-gated, Proposed and Future remain distinct.

Executive summary

Cerebro is a real, internally exercised architecture and verification estate - not yet a repeatable externally validated platform.

Cerebro Cyber Solutions is building a general-purpose AI operations platform as intellectual property. What exists today, and what this paper audits, is a real, internally exercised architecture and verification estate, not yet a repeatable, externally validated platform, in which governance, verification and provenance are properties of the architecture rather than a compliance layer added afterward. That estate has been exercised across operations, products, business development and content through roughly four months of versioned repository activity, from the first tracked commit on 18 April 2026 to this evidence cut on 14 August 2026.

The governance correction is as important as the technical work. Verified Completion still requires an independent checker whose result can stop a material transition. New verification work must now also name the bid, model, sellable artifact or live operator capability it protects. Controls that protect only more verification machinery default to accepted risk unless the operator overrides.

EXECUTIVE VERDICT / Cerebro has real internally exercised infrastructure and unusually explicit evidence controls. Its strongest proof is repository assurance, operational memory, one specialist model lane and one observed operator seam. Its principal gaps are end-to-end execution, adversarial containment, recovery drills, repeatable deployment and external outcomes.

The architecture

INTERFACE	Chat, Telegram, Command Center, documents and operator controls
AUTOMATION	Skills, agents, schedules, hooks, event triggers and governed handoffs
EXECUTION	Owned local models, specialist seats, tools and deterministic services
REASONING	Tiered local and cloud judgment under consent, routing and cost policy
MEMORY	File identity, human-readable knowledge and federated warehouse recall
OPERATOR + GOVERNANCE + VERIFIED COMPLETION span every layer	

Figure 1. The intended customer-facing five-layer design frame, governed as one stack.

The five-layer frame describes the intended customer experience: memory, reasoning, execution, automation and interface under operator authority. Internally, Brain, Skills, Agent Runtime and Fleet remain separate. Source presence is not runtime presence: a dated ledger observed the Composer Telegram seam, while broader Constellation modules were code-present and not running or parked. Three named positions divide the runtime's labor: Composer (orchestration and the observed operator seam), Sentinel (verification, evidence and gates) and Scholar (contained, fail-closed research, internally benchmarked in July 2026 as stronger on containment and weaker on iteration than five published research-agent designs). Only Composer's Telegram seam is recorded operating; Sentinel and Scholar are source-built, probe-qualified position descriptions, not live-agent claims.

The operating path includes one permanent, disclosed executor seat, a pinned build of the open-source Hermes agent runtime, under a ratified Self-Sufficiency exception; Cerebro owns the surrounding seam and control plane, and locally hosted models on owned hardware carry the workhorse load while cloud judgment remains an explicit, consented tier under routing and cost policy. Profiles and allowlists are identity and policy controls, not OS-level isolation. The estate does not yet provide platform-grade swarm isolation or hostile-model egress containment.

Routing is tiered like an organization: frontier judgment for the hardest calls, a senior-advisor tier, mid orchestration rungs and high-throughput small specialists, with the tiers enforced in the routing ledger as named rungs bound to explicit models.

What is materially new

Capability	Version 2.2 evidence
Governance	Ten adopted principles; Verified Completion is risk-tiered, and new verification must name the product asset it protects
Memory	File/local federation plus four default Qdrant collections, operational history, bounded reranking and fail-visible vector repair

Capability	Version 2.2 evidence
Model factory	One proven operating cyber-triage lane served by a Cerebro-trained model; one internally measured text-only trained artifact stored but not consumer-bound; four registered lane contracts; transfer-gate promotions graded by a ratified independent judge
Sentinel	Product definition and one served cyber-triage evidence line; no customer-deployment claim
Loop engineering	Bounded workers, verification and scheduled/event triggers exist; routing triad retired and broad orchestration not established
Framework currency	93/93 top-level estate entries rowed at the latest reconciliation, with available pin, license, review and disposition evidence and gaps stated; dated absorption decisions; patterns adopted, runtimes not surrendered
Model estate	Dated snapshot at the latest recorded probe: locally held ladder from a 4B training base to a 32B ratified judge, served trained specialist, embedding and rerank substrate, custody candidates held un-served; live residency probe-owned
Security	Accidental-exposure, path and provenance controls; no hostile-model containment or reliable namespaced-origin claim
Recovery	SQLite/database-dump and Qdrant capture evidence; multi-service restore remains partial
Business	Libro public alpha and BlackBox pre-ship; no repeatable deployment, external outcome or recurring-revenue proof claimed

Memory now includes operational kanban history as an eighth designed collection in Qdrant, the estate's vector database and a fourth default warehouse query lane. A measured cross-encoder change moved a real card-shaped target from rank 41 to 6 without broader-set regression. A later repair made oversized, malformed, non-finite, zero and wrong-dimension embeddings fail visibly. A dated internal run reported recall@5 of 0.957 with vector and rerank lanes healthy; it is an internal health shard, not an external benchmark.

Cybersecurity finding triage remains the only proven operating model-factory lane, and its served model is Cerebro's own, trained on owned hardware. A separate text-only Layer-0 artifact, also trained in the factory, scored 121/136 versus 104/136 for its documented base arm on an internal served-path comparison (paired McNemar $p=0.00297$), was stored with content-addressed evidence and remained unbound to any consumer. It is not a second production lane. Transfer-gate promotion verdicts are graded by a ratified, digest-pinned independent judge model rather than by the producing worker, and specialist seats bind to models through versioned route contracts whose ledger already records dated seat swaps as stronger admissible models arrived.

The factory behind those results is not thin. At this freeze its cyber exam chain holds 582 held-out cases across eight standing suites plus a separate promotion key; its documented run history, recorded in the factory resource ledger, roadmap and available per-run artifacts, includes a preference-optimization attempt that washed out and a seven-variant reinforcement-learning sweep recorded as exhausted; and its judge was ratified through a bake-off whose rejections, including a veto-winning candidate disqualified by a non-commercial license, are preserved in a canonical matrix. At the latest recorded residency probe before the freeze, the local estate spanned a locally held ladder from a 4-billion-parameter training base through 12, 26 and 31-billion-parameter seat models to a 32-billion-parameter ratified judge, alongside the served trained specialist, an embedding and reranking substrate, and custody candidates held un-served; live residency remains probe-owned.

The governance model

Principle set	Executive meaning
Governance + Parity + Lockstep	Authority, implementation and rollups stay aligned
Human-in-the-Mix + Least-Privilege	Automation stops before relationships, external commitments, representations and unauthorized scope
Reversibility + Observability	Changes preserve rollback and degradation becomes visible
Reproducibility + Self-Sufficiency	Inputs and transitions are replayable; one permanent disclosed executor seat is the narrow exception to native-runtime ownership
Verified Completion	Independent evidence decides when a material transition may proceed

Repository assurance now includes fourteen registered verification lanes (nine blocking, five advisory), load-bearing branch/path enforcement, source-generated close manifests with a sealed last-close pointer, passing-receipt-only reconciliation, a shared serialized receipt sink, and a client-side merge gate that holds each trunk merge until the authoritative remote test concludes green. An August hardening interval rebuilt the credential-exposure guard - four external adversarial re-gate rounds plus a final internal battery independently ratified by the decision layer - closing the original live leak vector and the known residual findings, after which the guard was frozen under a block-side-only precision invariant with further hardening admission-gated. The interval also added an advisory branch-drift detector that flags silent weakening of tracked security surfaces - detection, not commit-blocking - and completed the close-to-open seam with a pre-commit-owned seal and measurable carry-completeness. These controls strengthen repository truth. They do not prove customer outcomes, universal review or tamper-proof enforcement. The earlier routing triad was retired after producing no real council runs; its heuristic survives as doctrine rather than an operating orchestration claim.

Business and delivery state

Lane	Recorded state at evidence freeze
Business Development	Active capability and evidence packaging; external representation and commitment remain operator-owned

Lane	Recorded state at evidence freeze
Products	Libro v0.2.1-alpha in a separate public repository; BlackBox pre-ship; stable release remains owner-gated
Content	Distribution capability exists; no post-v2.1 external outcome upgrade recorded
Cyber Services	Dormant pending activation; offer materials do not equal an active client
Training	Deck and attendee bundles ready; no cohort or commercial delivery recorded
Finance	Dormant-in-fact and stale pending owner input; no guessed burn, revenue or weighted pipeline

Libro remains a public v0.2.1-alpha pre-release and BlackBox remains pre-ship. Business-development and content systems exist, but the repository contains no claim-grade proof of a repeatable customer deployment, externally validated outcome or recurring revenue. The next proof is one narrow, customer-reconstructable capability - not more horizontal infrastructure.

COMMERCIAL BOUNDARY / Internal build evidence is not repeatable customer deployment. A public alpha is not a stable release. A bounded internal benchmark is not external validation. Pipeline artifacts are not customer outcomes or recurring revenue.

Decision and engagement lens

For a regulated or security-sensitive buyer, the near-term question is whether one bounded capability can be reconstructed on buyer-controlled infrastructure with inspectable inputs, acceptance criteria, rollback and human approval. For a technology leader, the question is whether the discipline transfers without importing the entire internal estate. For a partner, the question is whether each claim is falsifiable and evidence supported.

Cerebro's differentiation is disciplined evidence across memory, models, workflows and operator authority. Its next value milestone is an independently accepted outcome, not another internal mechanism.